Unsupervised Learning

Course of Statistical Learning

Gareth et al.. An Introduction to Statistical Learning, Ch. 12

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Presentation Outline

Refresher: Supervised, Unsupervised, Reinforcement Learning

Differences and similarities

- Supervised learning: $\underline{y} = f(\underline{x})$
  - given $(\underline{x}, \underline{y})$ learn $f()$ [Function approximation]
- *Unsupervised learning*: $f(\underline{x})$
  - given $(\underline{x})$ learn $f()$ as a **compact representation** of $\underline{x}$ [Clustering]
- Reinforcement learning: $y = f(\underline{x}), \underline{z}$
  - given $(\underline{x}, \underline{z})$ learn $f()$ that generates y
  - $\underline{x}$ is the state, $\underline{z}$ is the reward signal

Challenges of Unsupervised Learning

- Usually **more challenging** than supervised learning
  - **more subjective**, not clearly defined goals
  - **difficult to assess results** obtained: no universally accepted mechanism to perform cross-validation
- Main issue: we do not have the labels $\Rightarrow$ we do **not have the ground truth**
- Several important **applications**:
  - find **similarities in genes** or **breast cancer samples** (or both!) to better understand the disease
  - find **similarities in online shopping/on demand platform users** for targeted advertisement
  - find **similar behaviors in in malware samples** to devise appropriate countermeasures

Presentation Outline

Refresher: Principal Component Analysis

- Can **summarize** a set of highly **correlated variables** with a smaller number of **representative** variables
- Reminder, **principal component directions**: **maximize variability** of data in the feature space
- Reminder, principal component directions: **as close as possible to the data**
- PCA
  - **methods** to compute the **principal components**
  - use of these components in the analysis of data
- PCA is an **unsupervised** approach, no associated responses
- Very useful for **data visualization** (particularly for high dimensional spaces)

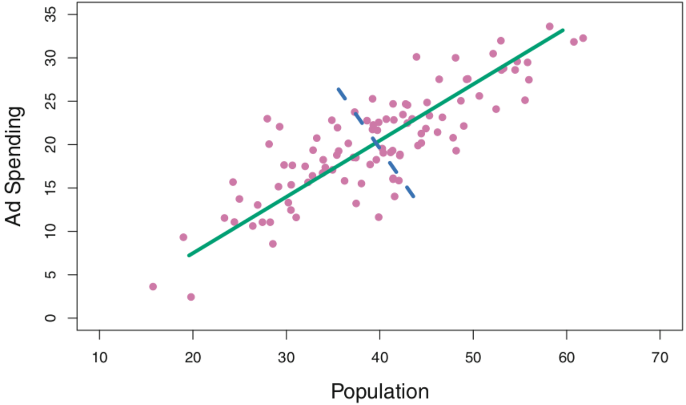
Principal Component Analysis

- Given a set of features $\{X_1, X_2, \ldots, X_p\}$
- **First Principal Component** Z_1 is the **normalized** linear combination of features that has the **largest variance**

$$Z_1 = \phi_{11}X_1 + \phi_{21}X_2 + \cdots + \phi_{p1}X_p \quad \text{where,} \quad \sum_{j=1}^{p} \phi_{j1}^2 = 1$$

- We refer to $\phi_1 = \{\phi_{11}, \phi_{21}, \ldots, \phi_{p1}\}^T$ as the **loading vector**
- We need to **constrain the loadings** (sum of squares equal to one) otherwise we can just increase their value to increase variance

Principal Component Analysis: visualization



Population size (pop) and spending (ad) for 100 different cities. Green line is principal component, blue dashed line second principal component, source [ISL]

Principal Component Analysis: computation

- Matrix $\mathbf{X}$, n rows (observations) and p columns (features)
- We care only about **variance**, we set the **mean of each column to zero**
- Look for **linear combination** of the sample features so to **maximize the variance** of the **principal component scores**

$$z_{i1} = \phi_{11}x_{i1} + \phi_{21}x_{i2} + \cdots + \phi_{p1}x_{ip} \quad \text{where,} \quad \sum_{j=1}^{p} \phi_{j1}^2 = 1$$

- Since each X_j has zero mean so does Z_1, hence the **variance** is $\frac{1}{n}\sum_{i=1}^{n} z_{i1}^2$
- Then we solve the following **optimization problem**:

$$\max_{\phi_{11},\phi_{21},\ldots,\phi_{p1}} \frac{1}{n}\sum_{i=1}^{n}(\phi_{11}x_{i1} + \phi_{21}x_{i2} + \cdots + \phi_{p1}x_{ip})^2 \quad \text{subject to} \quad \sum_{j=1}^{p} \phi_{j1}^2 = 1$$

Further Principal Components

- The **second principal component** is the linear combination of $X_1, X_2, \ldots, X_p$ that has **maximal variance** and is **uncorrelated** with Z_1

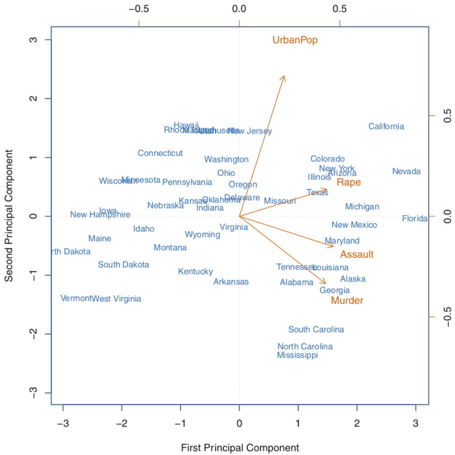
$$z_{i2} = \phi_{12}x_{i1} + \phi_{22}x_{i2} + \cdots + \phi_{p2}x_{ip} \quad \text{where,} \quad \sum_{j=1}^{p} \phi_{j2}^2 = 1$$

- Constraining Z_2 to be **uncorrelated** with Z_1 is equivalent to constrain the direction of ϕ_2 to be **othogonal** to ϕ_1, and so forth
- This problem can be solved using **Singular Value Decomposition** of the matrix $\mathbf{X}$
- It turns out that the **principal component directions** $\phi_1, \phi_2, \ldots, \phi_K$ are the ordered sequence of the *eigenvectors* of the matrix $\mathbf{X}^T\mathbf{X}$ while the **variance** of the components are the **eigenvalues**. $K = \min\{n-1, p\}$

USA arrest dataset

- **USA arrest** data
  - For each of the **50 states** in U.S. record number of arrests every $100,000$ residents
  - **Arrests** are divided in three categories (**features**): **Assault**, **Murder** and **Rape**
  - Data contains a **fourth feature** which is the **UrbanPop**: percentage of population living in urban areas
- Number of points $n = 50$ (score vector), features $p = 4$ (loading vectors)
- PCA performed after **data standardization** (zero means, standard deviation of one)

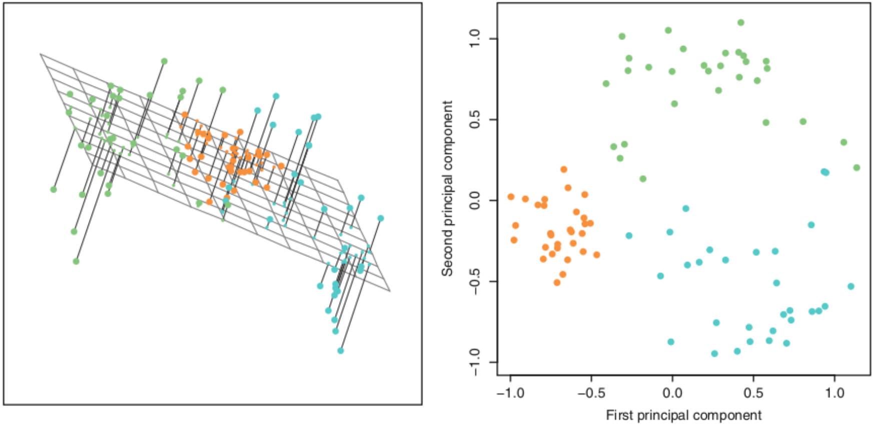
PCA for USA arrest



	PC1	PC2
Murder	0.5358995	−0.4181809
Assault	0.5831836	−0.1879856
UrbanPop	0.2781909	0.8728062
Rape	0.5434321	0.1673186

First two principal components for the USAarrest data, blue name represent scores for the corresponding state, orange arrows indicate loading vectors for the two first principal components (values are on right and top of the axis). Source [ISL]

Another interpretation for PCA: visualization



Simulated observations (90) in three dimensions. The first two principal components identifies the plane that *best fits* the data. Source [ISL]

PCA as closest linear surface to the observations

- Basic interpretation: **loading vectors** are the directions in feature space along which data **vary the most**
- Other interpretation: low-dimensional linear surfaces that are **closest to the observations**
- First Principal Component
  - line in p-dimensional space that is **closest to the n observations**
  - closest according to **average squared Euclidean distance**
  - single dimension that **best approximates** all the observations
- First **M Principal Components** and the associated **score vectors** provide the best **M-dimensional approximation** to the i-th observation x_{ij}

$$x_{ij} \approx \sum_{m=1}^{M} z_{im}\phi_{jm}$$

Proportion of Variance Explained I

- **Strength** of a **component** $\Rightarrow$ **Proportion of Variance Explained (PVE)** by that component
- **Total variance** in a data set assuming data have been centered to have zero mean:

$$\sum_{j=1}^{p} Var(X_j) = \sum_{j=1}^{p} \frac{1}{n} \sum_{i=1}^{n} x_{ij}^2$$

- **Variance** explained by m**-th principal component**

$$Var(Z_m) = \frac{1}{n} \sum_{i=1}^{n} z_{im}^2$$

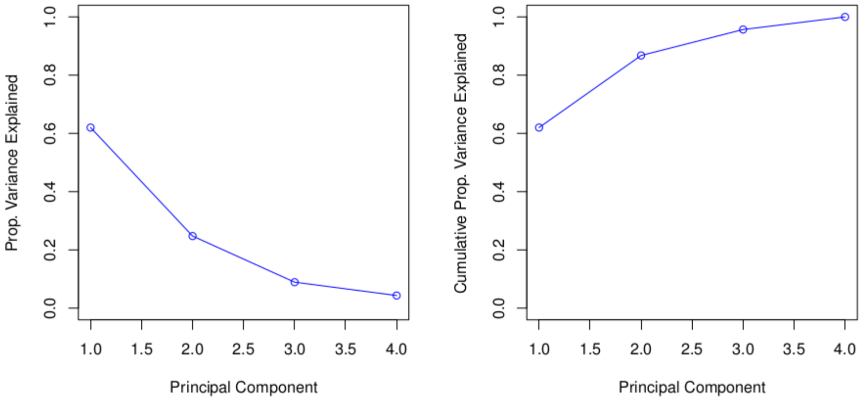
- It can be shown that when $M = \min\{n-1, p\}$ then

$$\sum_{j=1}^{p} Var(X_j) = \sum_{m=1}^{M} Var(Z_m)$$

Proportion of Variance Explained II

PVE is defined as

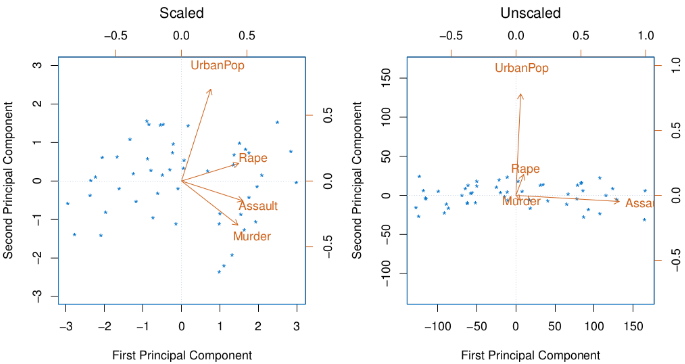
$$\frac{\sum_{i=1}^{n} z_{im}^2}{\sum_{j=1}^{p} \sum_{i=1}^{n} x_{ij}^2}$$



How Many Principal Components Should we Use

- **How many components** do we **need** to have a **good representation** of our data? No simple answer, we can not use cross-validation
  - Why not?
  - When could we use Cross-Validation?
- The "scree" plot can be used, we look for an **elbow**

Scaling the variables



If variables are in **different units** it is **important to scale** them to have standard deviation equal to one. Source [ISL]

Presentation Outline

Clustering

- Broad set of **techniques** to find **subgroups** (also called **clusters**) in a data set
- **Partition** of the data set in groups so that observations within each group are **similar** to each other (and different from observations in other groups)
- Key point: specify what we mean by **similar** and **different**
- **Similarity** (or **dissimilarity**) is often a domain specific concept, based on knowledge of the data
- **PCA vs Clustering**
  - **PCA** looks for a **low-dimensional representation** of the observations that explains a large fraction of the variance
  - **Clustering** looks for **homogenous subgroups** among the observations

Example: Market Segmentation for targeted advertising

- Large number of measurements for a large number of **people**
  - median household income, occupation, distance from nearest urban area, ...
- **Identify subgroups of people** that may be more **receptive** to a particular form of **advertising** or that may be more interested in a given **product**
- **Market segmentation** $\Rightarrow$ Clustering people in the data set

Two clustering methods

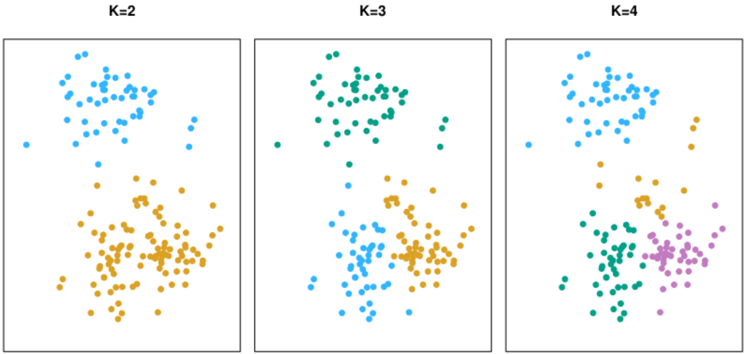
K-means clustering:

- Partition observation into a **predefined number of clusters** (k)

Hierarchical clustering

- Build a **tree-like representation** of the observation, **dendrogram**
- Dendrogram: allows to visualize the clusterings **for each possible number of clusters**, $1, \ldots, n$

K-means clustering



Simulated data set with 150 observations in 2-dimensional space. Panels show the **outcome** of a K-means clustering algorithm for $K = 2, 3, 4$. **Colors** represent the outcome of the clustering algorithm **not labels**, source [ISL]

Details of K-means clustering I

- $C_1, \ldots, C_K$ are a **partition** of the indices of the observations
- If **observation** i belongs to **cluster** k then $i \in C_k$
  - Each **observation** belongs to **at least one cluster**

$$\bigcup_{i=1}^{K} C_i = \{1, 2, \ldots, n\}$$

  - Clusters are **non-overlapping** (no observation belongs to more than one cluster)

$$\forall k, k' | k \neq k', C_k \cap C_{k'} = \emptyset$$

Details of K-means clustering

- **Within-Cluster Variation** $WCV(C_k)$ measures how much the observations that belong to **cluster** C_k **differ from each other**

$$\arg\min_{C_1,\ldots,C_K}\left\{\sum_{k=1}^{K} WCV(C_k)\right\}$$

where $WCV(C_k)$ depends on application. A popular choice is the **Squared Euclidean Distance**

$$WCV(C_k) = \frac{1}{|C_k|}\sum_{i,i'\in C_k}\sum_{j=1}^{p}(x_{ij}-x_{i'j})^2$$

- **Objective function** for the optimization problem that defines **K-means clustering**

$$\arg\min_{C_1,\ldots,C_K}\left\{\sum_{k=1}^{K}\frac{1}{|C_k|}\sum_{i,i'\in C_k}\sum_{j=1}^{p}(x_{ij}-x_{i'j})^2\right\}$$

K-means Clustering Algorithm

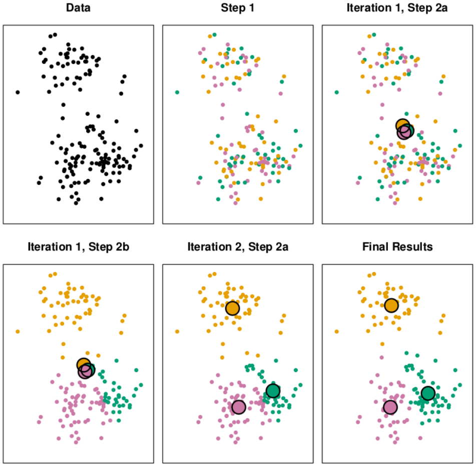
1. Randomly **assign a number** from $1, \ldots, K$ to each of the **observations**. This is the **initial cluster assignment**

2. **Iterate** until convergence (i.e., the cluster assignment does not change)

   2.1 For each cluster **compute the cluster centroid**, i.e., vector of p feature means for the observations in the k-th cluster

   2.2 **Assign** each **observation** to the **closest centroid**, using Euclidean Distance.

K-means clustering: example



Visualization of the iterations for the K-means, source [ISL]

Properties of K-means Clustering Algorithm

- The **K-means** clustering algorithm is **guaranteed** to **decrease** the value of the **objective function** at **each step**, note that:

$$\frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=i}^{p} (x_{ij} - x_{i'j})^2 = 2 \sum_{i \in C_k} \sum_{j=i}^{p} (x_{ij} - \bar{x}_{kj})^2$$

where, $\bar{x}_{kj} = \frac{1}{|C_k|} \sum_{i \in C_k} x_{ij}$ is the **mean** for feature j in **cluster** C_k

- The algorithm is **not guaranteed** to find the **global optimum**

K-means clustering: different initializations

Results with different initializations for the K-means, source [ISL]

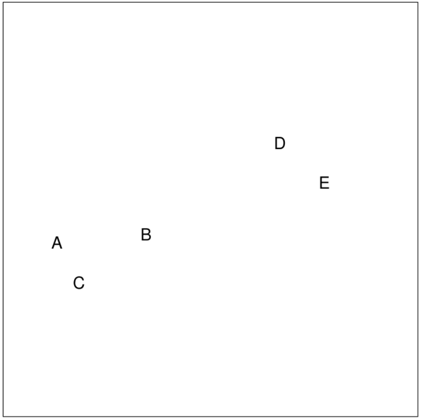
Presentation Outline

Hierarchical Clustering

- Builds and maintains a **hierarchy of clusters**
- **Bottom-up** or **agglomerative** clustering
  - A **dendogram** is built **from leaves** (i.e., observations) combining clusters up to the trunk
  - Most common type of **hierarchical clustering**

Hierarchical Clustering, overall idea

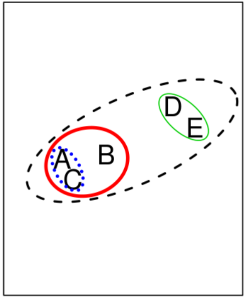
Main idea: to build a hierarchy in a bottom-up fashion

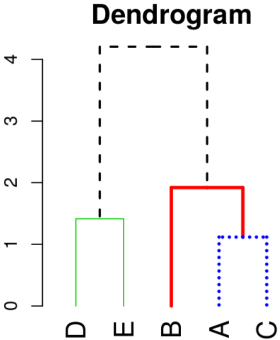


courtesy of T. Hastie and R. Tibshirani

Hierarchical Clustering Algorithm

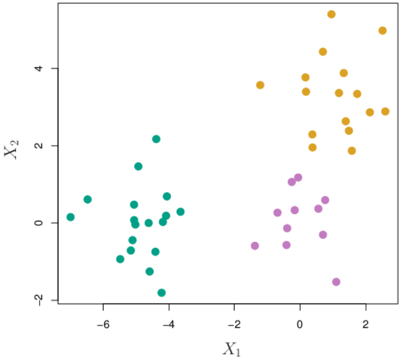
1. Start with **each point** in its own **cluster**
2. Identify the **closest couple of clusters** and **merge** them
3. Repeat from 2 until all points are in a single cluster





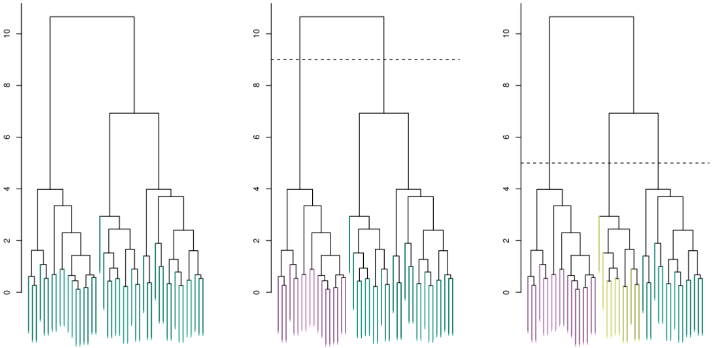
courtesy of T. Hastie and R. Tibshirani

Hierarchical Clustering, Example



45 observations generated in 2-dimensional space. Colors represent three classes that are unknown to the clustering approach. Goal is to cluster observations and discover classes from data, source [ISL]

Hierarchical Clustering, Results

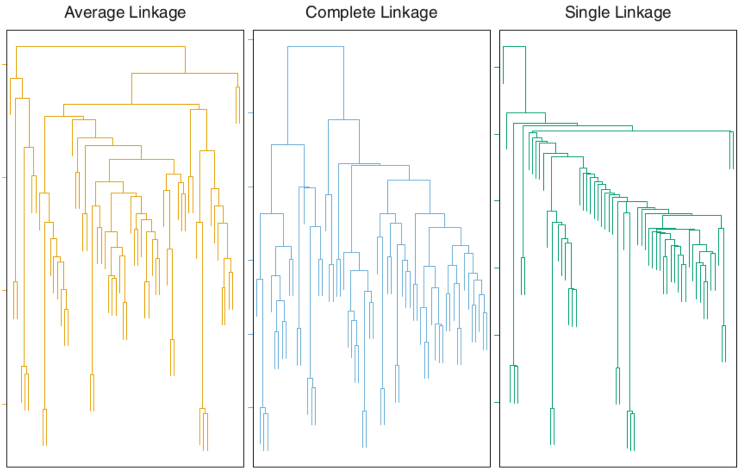


Left, dendogram for hierarchical clustering using complete linkage and Euclidean distance; Center, same dendogram cut at a height of 9, this results in two distinct clusters (different colors); Right, same dendogram cut at a height of 5, this results in three distinct clusters (different colors), source [ISL]

Types of Linkage

Linkage	Description
Complete	**Maximal** inter-cluster **dissimilarity**. Compute **all pairwise** dissimilarities between observations in cluster A and the observations in cluster B, record the **largest** dissimilarities
Single	**Minimal** inter-cluster dissimilarity. Compute all pairwise dissimilarity between observations in cluster A and the observations in cluster B, record the **smallest** of these dissimilarities.
Average	**Mean** inter-cluster dissimilarity. Compute all pairwise dissimilarity between observations in cluster A and the observations in cluster B, record the **average** of these dissimilarities.
Centroids	Dissimilarity between the **centroid** for cluster A (a mean vector of length p) and the **centroid** in cluster B. Centroid linkage can result in undesirable **inversion**.

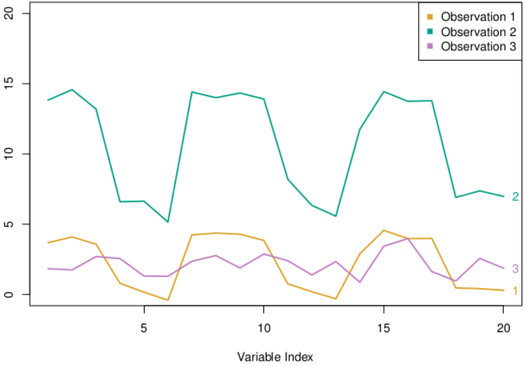
Comparison of different linkage types



Comparison of different linkage types, source [ISL]

Choice of Dissimilarity Measure

- **Euclidean distance** is typically used
- **Correlation** is a possible alternative: two observations are **similar** if they are highly **correlated**



Three observations with measurements on 20 variables, source [ISL]

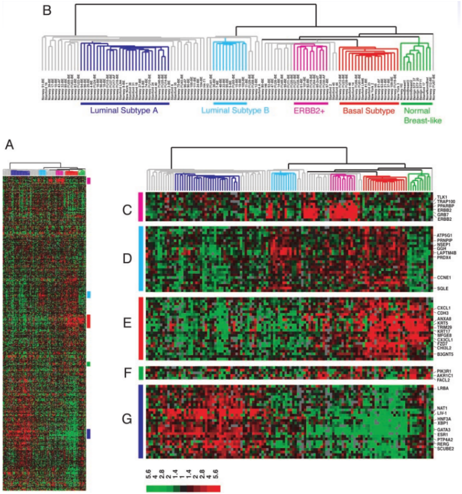
Practical Issues

- **Scaling of the variables** is important. Should we **standardize** observations before applying clustering?
- For hierarchical clustering
  - Which **dissimilarity measure**?
  - What **type of linkage** should be used?
- **How many clusters** should we choose?
- **Which features** should we use to drive the clustering?

Example: breast cancer microarray study I

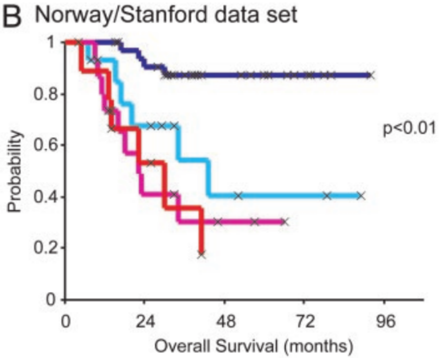
- **Repeated observation** of **breast tumor subtypes** in independent gene expression data sets, Sorlie et al., PNAS 2003.
- **Gene expression measurement** for about **8000 genes**, for each of **88 breast cancer patients**
- **Average** linkage, **correlation** metric
- Clustered samples using **500 intrinsic genes**: each woman was measured **before** and **after** chemiotherapy. Intrinsic genes have smallest within/between variation.

Example: breast cancer microarray study II



Sorlie et al. PNAS 2003

Example: breast cancer microarray study III



Sorlie et al. PNAS 2003

Conclusions

- **Unsupervised Learning** is important for understanding the **variation** and **structure** of an **unlabeled** dataset
- Can be a useful step **before** performing **supervised learning**
- Intrisically **more difficult** than **supervised learning**, no ground truth (i.e., label) and no single **objective** (i.e., test prediction accuracy)
- Challenging and **active field of research** with many recently developed tools: **self-organized maps**, **spectral clustering**, **affinity propagation**, and so forth.